On Layer Normalization in the Transformer Architecture

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Abstract

This paper investigates architectures with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding On Layer Normalization in the Transformer Architecture.

2. Technical Methodology & Empirical Findings

Section topic: Architectures
Question context: What is the difference between Pre-LN and Post-LN Transformer block configurations?
Key Finding: Pre-LN applies Layer Normalization before each sublayer rather than after the residual addition, preventing vanishing/exploding gradients at early iterations and stabilizing deep models [On Layer Normalization, p. 4].
Evidence passage: Post-LN applies LayerNorm after residual addition; Pre-LN applies LayerNorm inside residual branch before sublayer Pre-LN avoids gradient vanishing at initialization and enables warmup-free training

3. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for On Layer Normalization in the Transformer Architecture. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

4. Technical Methodology & Empirical Findings

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